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Answer: A

Horizontally,

$$u \cos 30^\circ t = 3.0 \Rightarrow t = \frac{3.0}{u \cos 30^\circ}$$

Vertically, taking upwards as positive,

$$u \sin 30^\circ t - \frac{1}{2}gt^2 = -0.25$$

Sub-in the expression for t above,

$$3.0 \frac{\sin 30^\circ}{\cos 30^\circ} - \frac{1}{2}g \left(\frac{3.0}{u \cos 30^\circ} \right)^2 = -0.25$$

Solving, $u^2 = 29.7$, $u = 5.4 \text{ m s}^{-1}$

Option C: forgot to take square root

Option B: took vertical displacement to be 0.25 m

Option D: took vertical displacement to be 0.25 m and forgot to take square root

